began to have trouble understanding his speech, I sometimes found myself saying over and over again, as many as ten times, “Stephen, I still don’t understand; please say it again.” Showing a bit of frustration, he would continue to repeat himself until I suddenly understood: He was telling me a wonderfully funny, off-the-wall, one-line joke. When I finally caught it, he would grin with pleasure.)

Entropy

Having extolled Hawking’s ability to out-think and out-intuit all his colleague-competitors, I must now confess that he has not managed to do so *all* the time, just most. Among his defeats, perhaps the most spectacular was at the hands of one of John Wheeler’s graduate students, Jacob Bekenstein. But in the midst of that defeat, as we shall see, Hawking produced a far greater triumph: his discovery that black holes can evaporate. The tortuous route to that discovery will occupy much of the rest of this chapter.

The playing field on which Hawking was defeated was that of *black-hole thermodynamics.* Thermodynamics is the set of physical laws that govern the random, statistical behavior of large numbers of atoms, for example, the atoms that make up the air in a room or those that make up the entire Sun. The atoms’ statistical behavior includes, among other things, their random jiggling caused by heat; and correspondingly, the laws of thermodynamics include, among other things, the laws that govern heat. Hence the name *thermodynamics.*

A year before Hawking discovered his area theorem, Demetrios Christodoulou, a nineteen-year-old graduate student in Wheeler’s Princeton group, noticed that the equations that describe slow changes in the properties of black holes (for example, when they slowly accrete gas) resemble some of the equations of thermodynamics. The resemblance was remarkable, but there was no reason to think it anything more than a coincidence.

This resemblance was strengthened by Hawking’s area theorem: The area theorem closely resembled the *second law of thermodynamics.* In fact, the area theorem, as expressed earlier in this chapter, becomes the second law of thermodynamics if we merely replace the phrase “horizon areas” by the word entropy: *In any region of space, and at any moment of time (as*